Assignment Topic: Computers in persuasive roles-Simulated cause-and-effect scenarios

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Simulation name

CarePath CMH: Hospital Visit Planning Simulator

Main purpose

To show how a patient's choices before visiting a hospital affect:

- waiting time
- correct department
- doctor availability
- walking route
- stress level
- chance of missing the doctor
- visit success

Screen 1: Welcome Screen

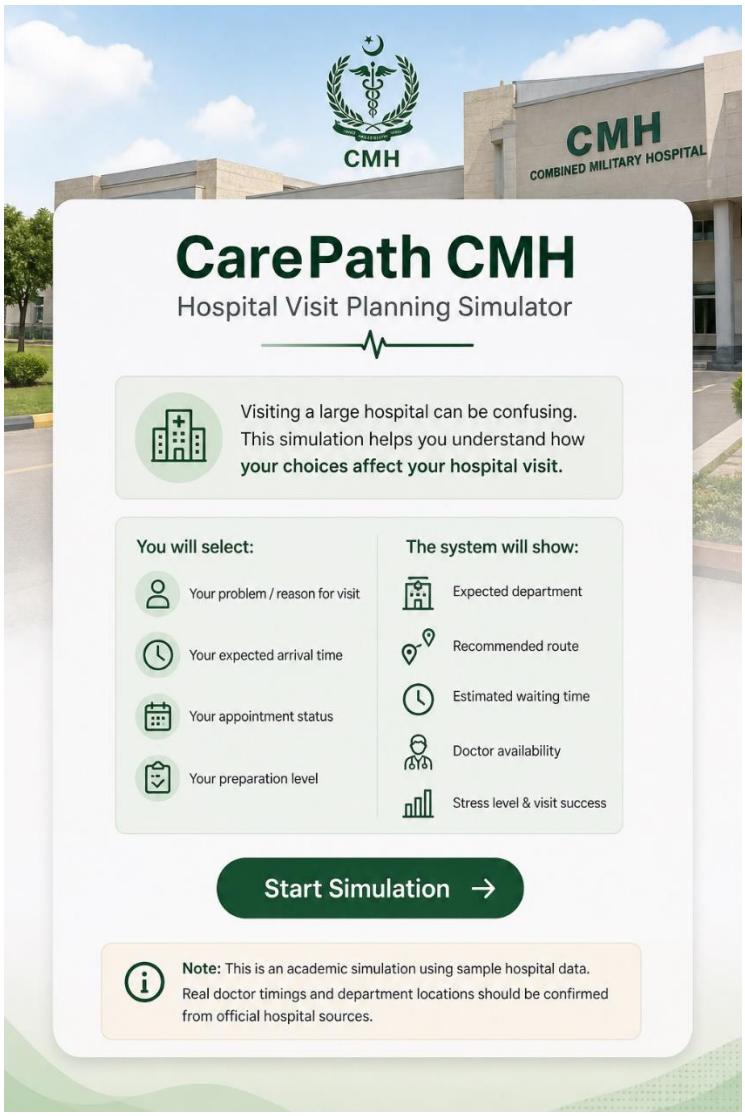
Purpose

This screen introduces the simulation and explains what the user will experience.

Important UI elements

Element	Purpose
App title	Gives identity
Short explanation	Tells user this is a simulation
Start button	Moves to form screen
Small disclaimer	Says data is simulated

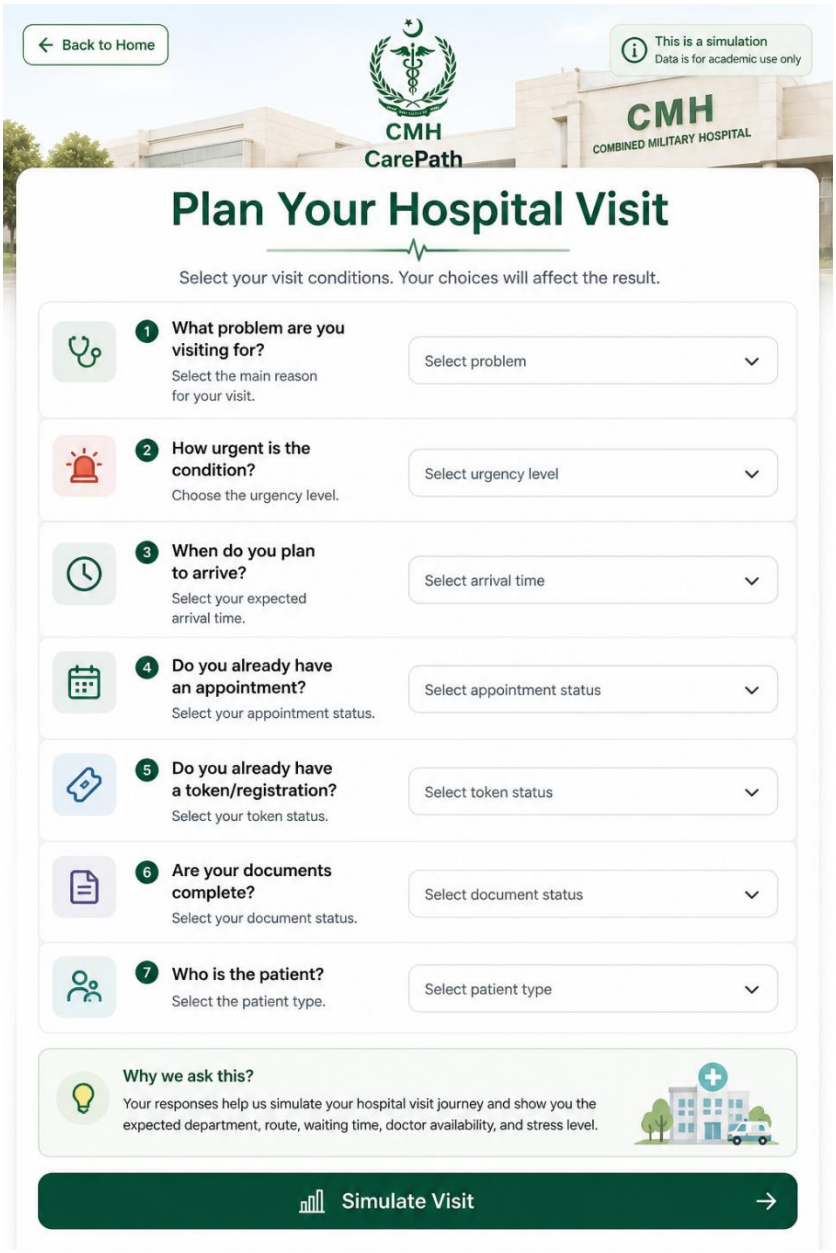
Screen:



Screen 2: Cause Form Screen

This is the most important screen.

Here the user selects all the **causes / independent variables**.



Form field 1: Patient problem

Label

What problem are you visiting for?

Dropdown options

Fever / weakness
Chest pain
Child illness
Bone or joint pain

Injury / fracture
Eye problem
Skin problem
Dental problem
Pregnancy-related issue
Follow-up checkup
Medicine refill
Lab test / report collection

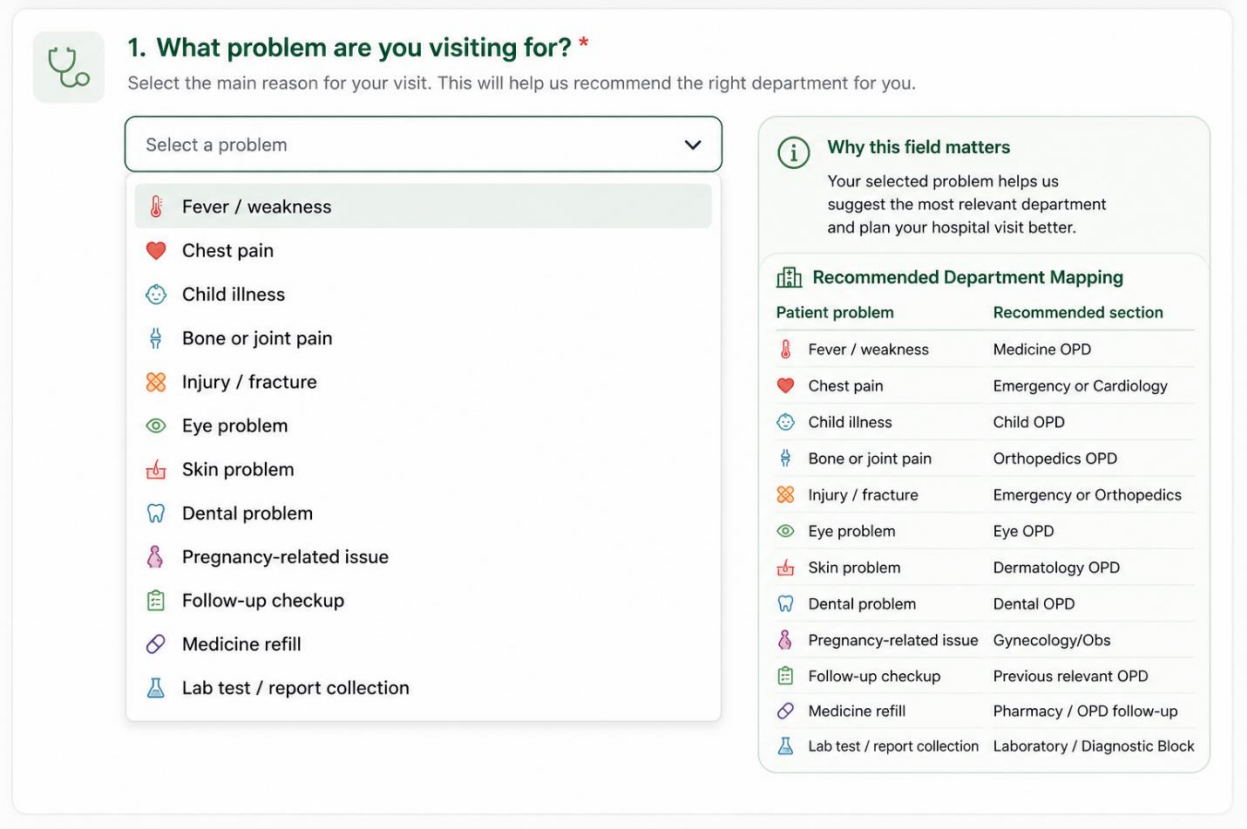
Why this field matters

This decides the **recommended department**.

Effect mapping

Patient problem	**Recommended section**
Fever / weakness	Medicine OPD
Chest pain	Emergency or Cardiology
Child illness	Child OPD
Bone or joint pain	Orthopedics OPD
Injury / fracture	Emergency or Orthopedics
Eye problem	Eye OPD
Skin problem	Dermatology OPD
Dental problem	Dental OPD
Pregnancy-related issue	Gynecology/Obs
Follow-up checkup	Previous relevant OPD
Medicine refill	Pharmacy / OPD follow-up
Lab test / report collection	Laboratory / Diagnostic Block

Patient problem **Recommended section**



Form field 2: Urgency level

Label

How urgent is the condition?

Dropdown options

Emergency
Urgent but stable
Routine OPD visit
Follow-up only

Why this field matters

This decides whether the patient should go to **Emergency** or **OPD**.

Effect mapping

Urgency	Effect
Emergency	Direct route to Emergency, OPD skipped
Urgent but stable	Warning shown, faster department suggestion
Routine OPD visit	Normal OPD route
Follow-up only	Shorter route, lower waiting time if appointment exists

Example system logic

If the user selects:

Problem: Chest pain
Urgency: Emergency

System response:

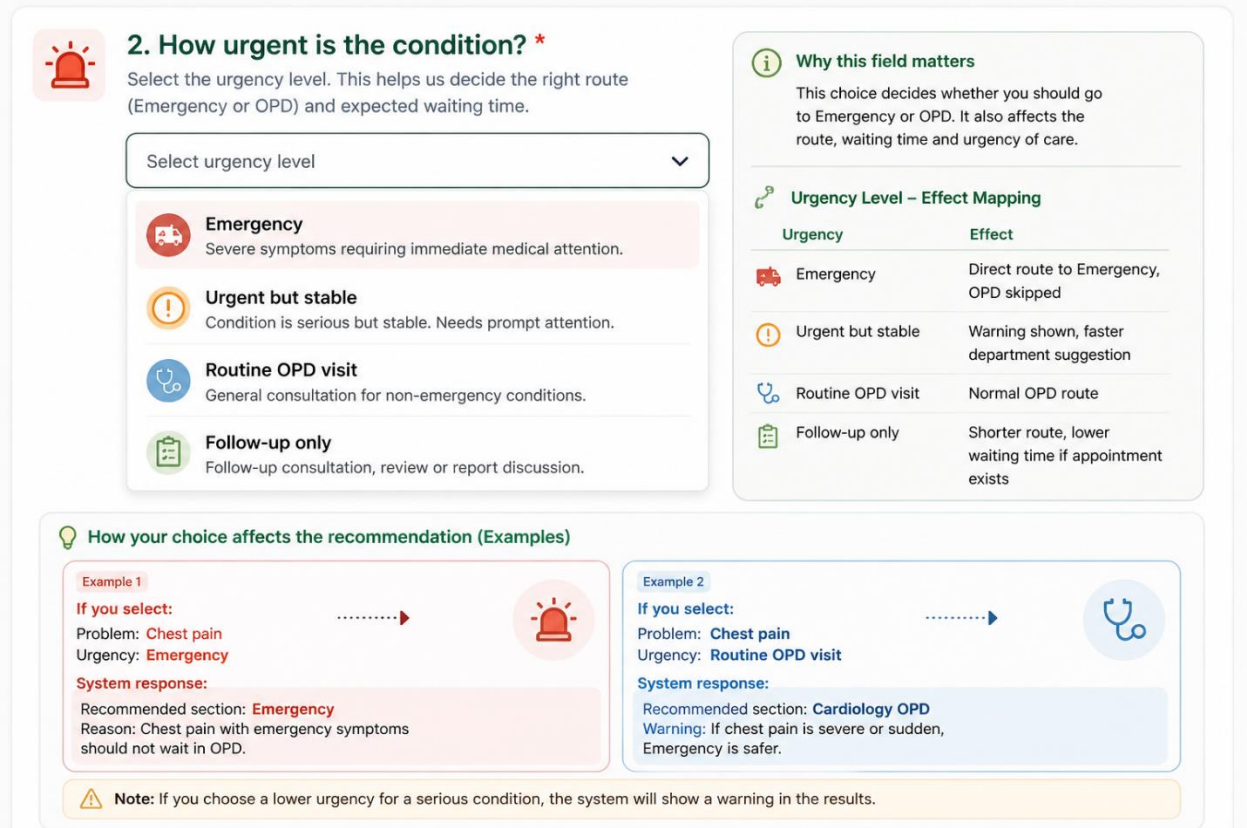
Recommended section: Emergency
Reason: Chest pain with emergency symptoms should not wait in OPD.

If the user selects:

Problem: Chest pain
Urgency: Routine OPD

System response:

Recommended section: Cardiology OPD
Warning: If chest pain is severe or sudden, Emergency is safer.



Form field 3: Arrival time

Label

When do you plan to arrive?

Dropdown options

8:00 AM - 9:00 AM
9:00 AM - 10:30 AM
10:30 AM - 12:00 PM
12:00 PM - 1:30 PM
After 1:30 PM

Why this field matters

This affects:

- waiting time
- doctor availability

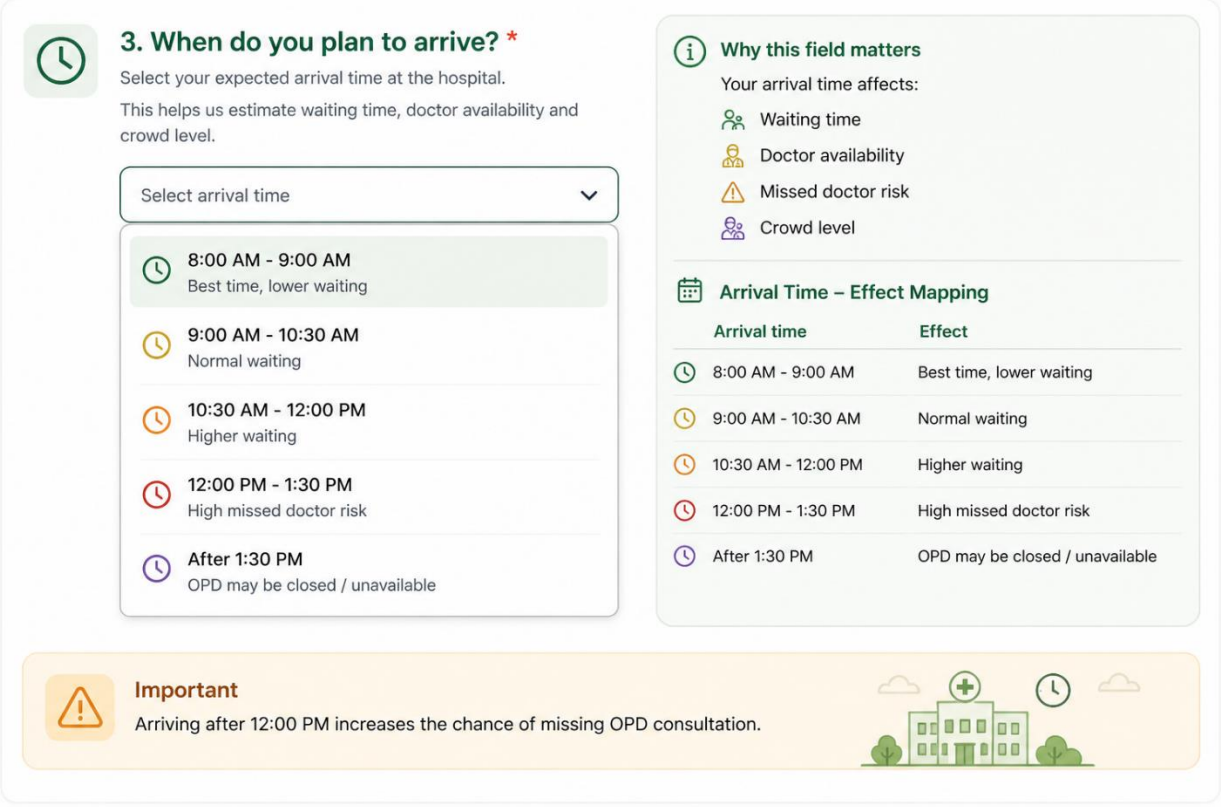
- missed doctor risk
- crowd level

Effect mapping

Arrival time	Effect
8:00 AM - 9:00 AM	Best time, lower waiting
9:00 AM - 10:30 AM	Normal waiting
10:30 AM - 12:00 PM	Higher waiting
12:00 PM - 1:30 PM	High missed doctor risk
After 1:30 PM	OPD may be closed/unavailable

Suggested result wording

Arriving after 12:00 PM increases the chance of missing OPD consultation.



Form field 4: Appointment status

Label

Do you already have an appointment?

Dropdown options

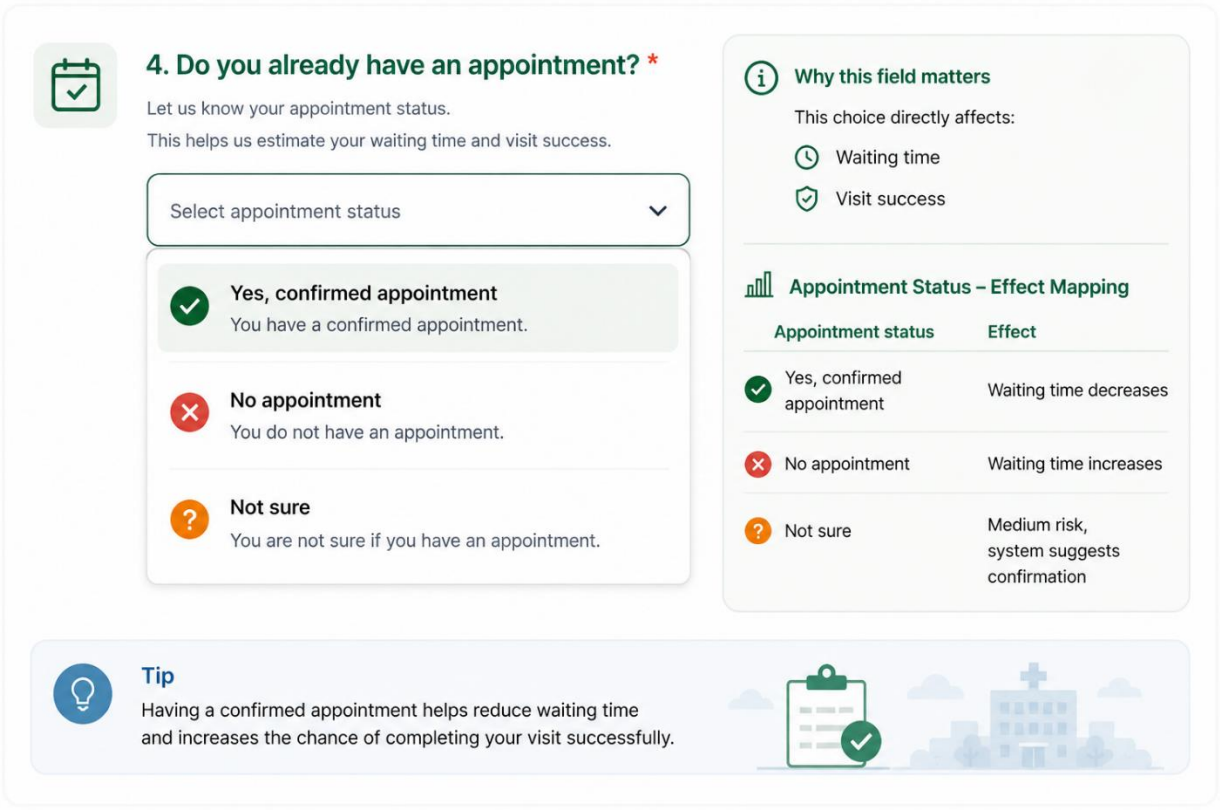
Yes, confirmed appointment
No appointment
Not sure

Why this field matters

This affects waiting time and visit success.

Effect mapping

Appointment status	Effect
Yes, confirmed appointment	Waiting time decreases
No appointment	Waiting time increases
Not sure	Medium risk, system suggests confirmation



Form field 5: Token status

Label

Do you already have a token/registration?

Dropdown options

Yes, token already taken
No, need token at hospital
Not applicable

Why this field matters

This affects the route and waiting time.

Effect mapping

Token status	Effect
Yes, token already taken	Token counter skipped

Token status	Effect
No, need token at hospital	Token counter added to route
Not applicable	Used for emergency/lab/report cases

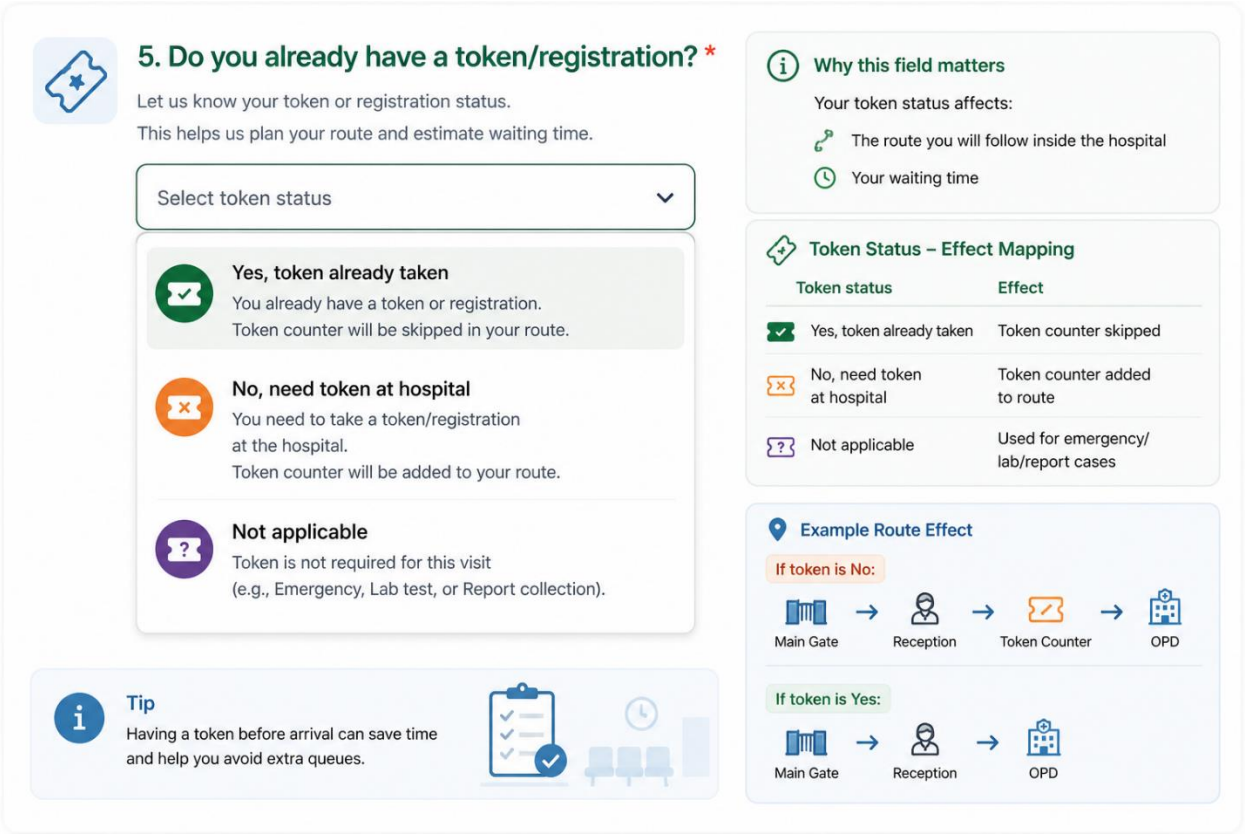
Example route effect

If token is **No**:

Main Gate → Reception → Token Counter → OPD

If token is **Yes**:

Main Gate → Reception → OPD



Form field 6: Documents status

Label

Are your documents complete?

Dropdown options

Complete documents
Incomplete documents
Not sure what to bring

Why this field matters

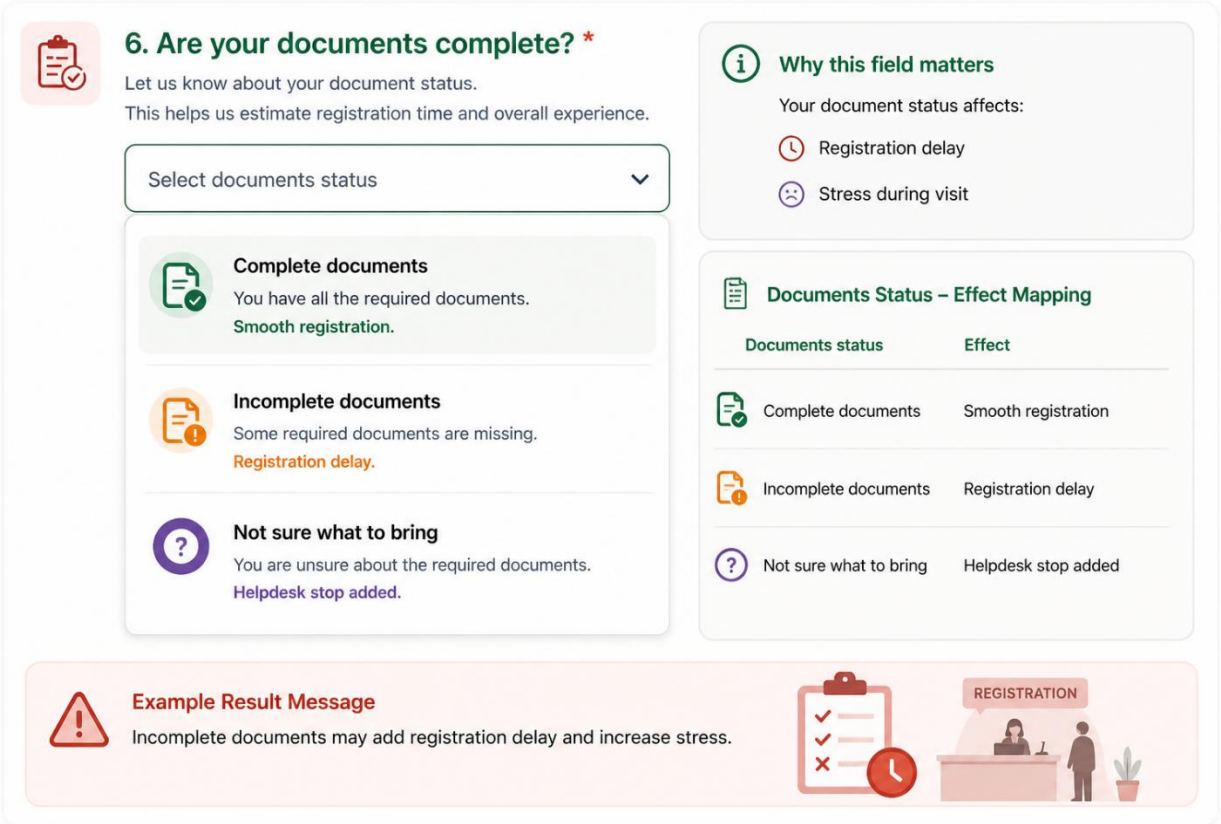
This affects registration delay and stress.

Effect mapping

Documents status	Effect
Complete documents	Smooth registration
Incomplete documents	Registration delay
Not sure what to bring	Helpdesk stop added

Example result message

Incomplete documents may add registration delay and increase stress.



Form field 7: Patient type

Label

Who is the patient?

Dropdown options

Adult, normal walking
Elderly patient
Child patient
Wheelchair user
Patient with attendant

Why this field matters

This affects walking effort and route difficulty.

Effect mapping

Patient type	Effect
Adult, normal walking	Normal route
Elderly patient	Walking effort increases
Child patient	Child-friendly department suggestion
Wheelchair user	Accessible route needed
Patient with attendant	Stress/confusion slightly reduced

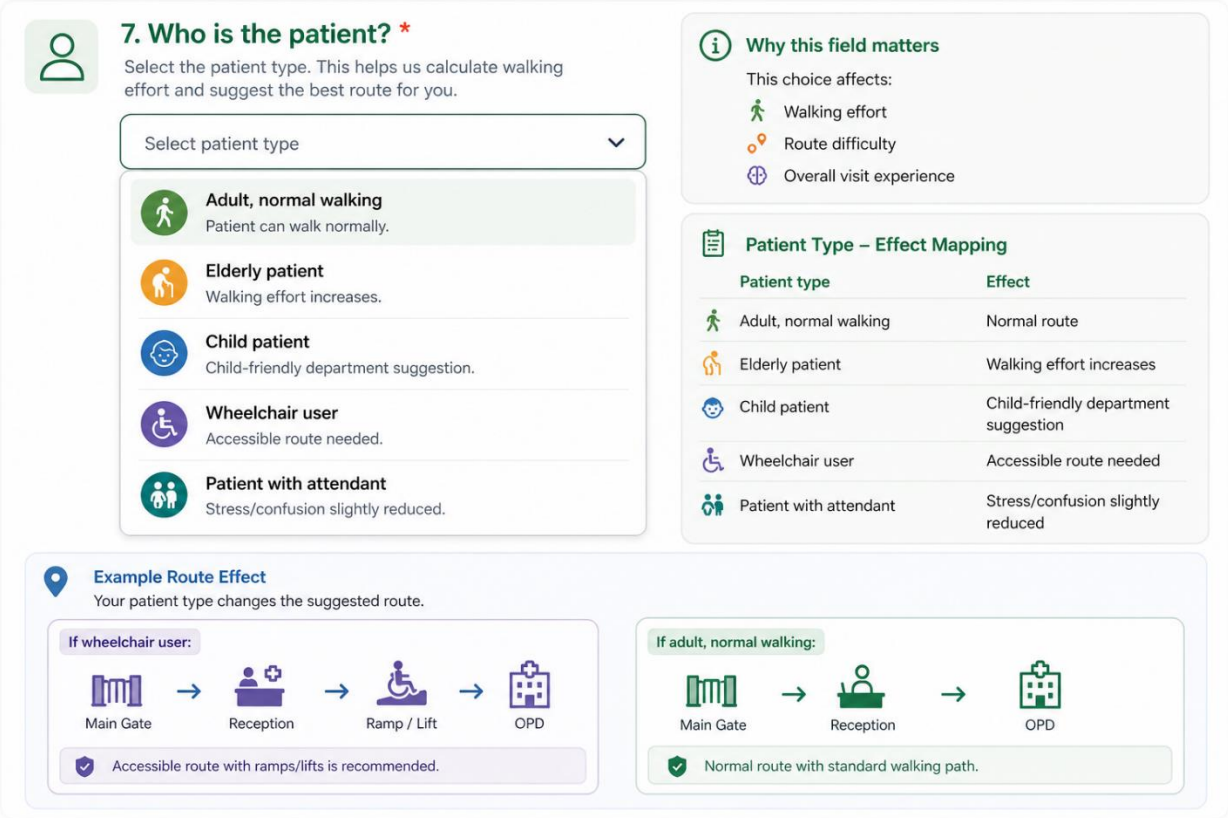
Example route effect

If wheelchair user:

Main Gate → Reception → Ramp/Lift → OPD

If adult normal walking:

Main Gate → Reception → OPD



Form field 8: First-time visitor

Label

Is this your first visit to this hospital?

Dropdown options

Yes
No

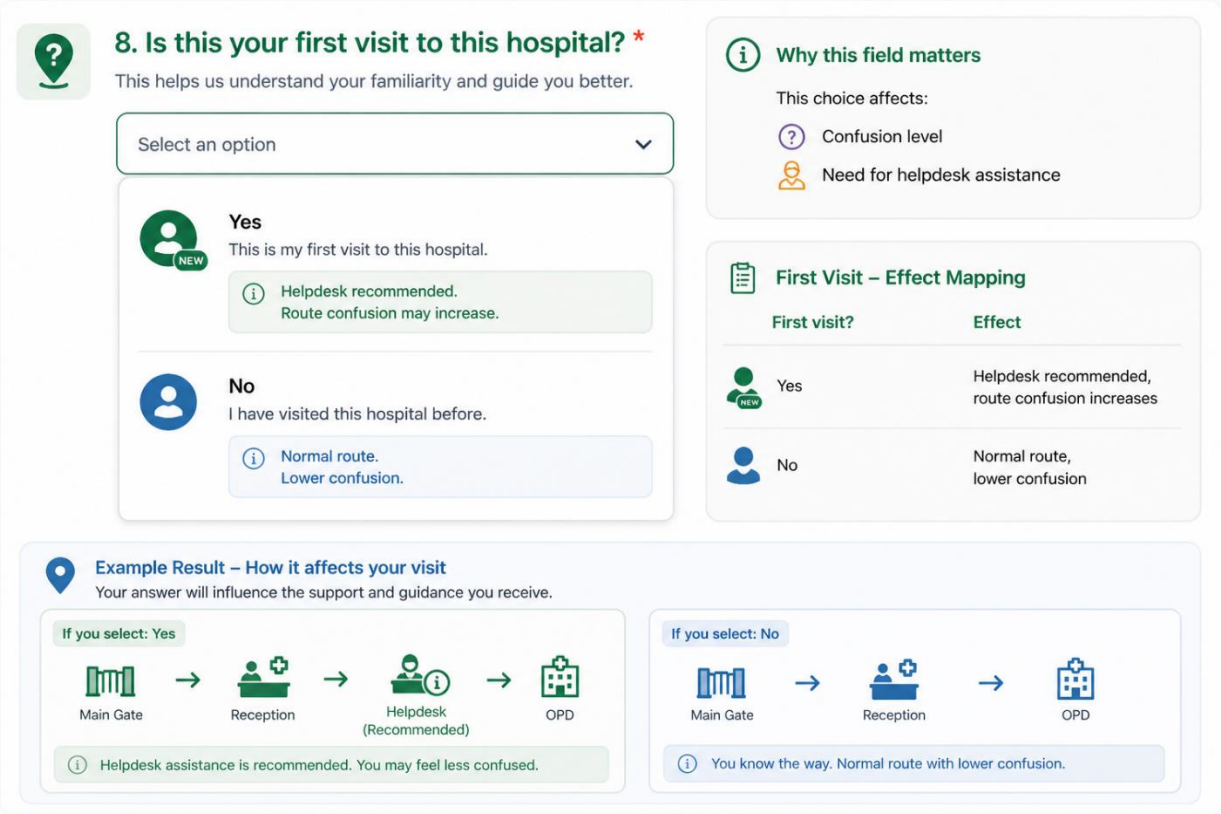
Why this field matters

This affects confusion level and need for helpdesk.

Effect mapping

First visit? Effect

Yes	Helpdesk recommended, route confusion increases
No	Normal route, lower confusion



Screen 3: Effect / System Response Screen

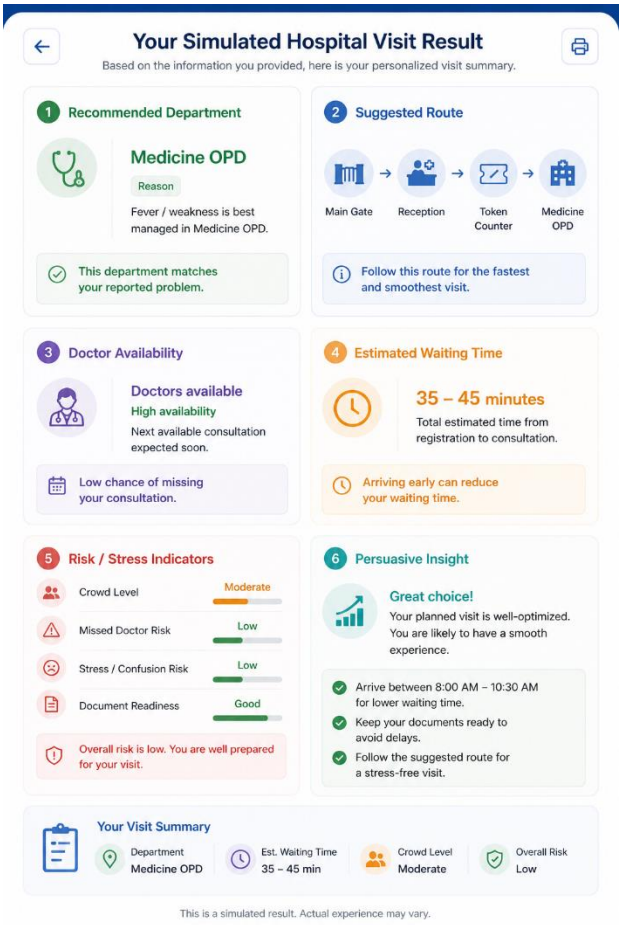
This is where the simulation shows results.

Main sections on this screen

The response screen should have **six result blocks**:

1. Recommended department
2. Suggested route
3. Doctor availability
4. Estimated waiting time
5. Risk/stress indicators

6. Persuasive insight



Result block 1: Recommended department

Example

Recommended Section:
Medicine OPD

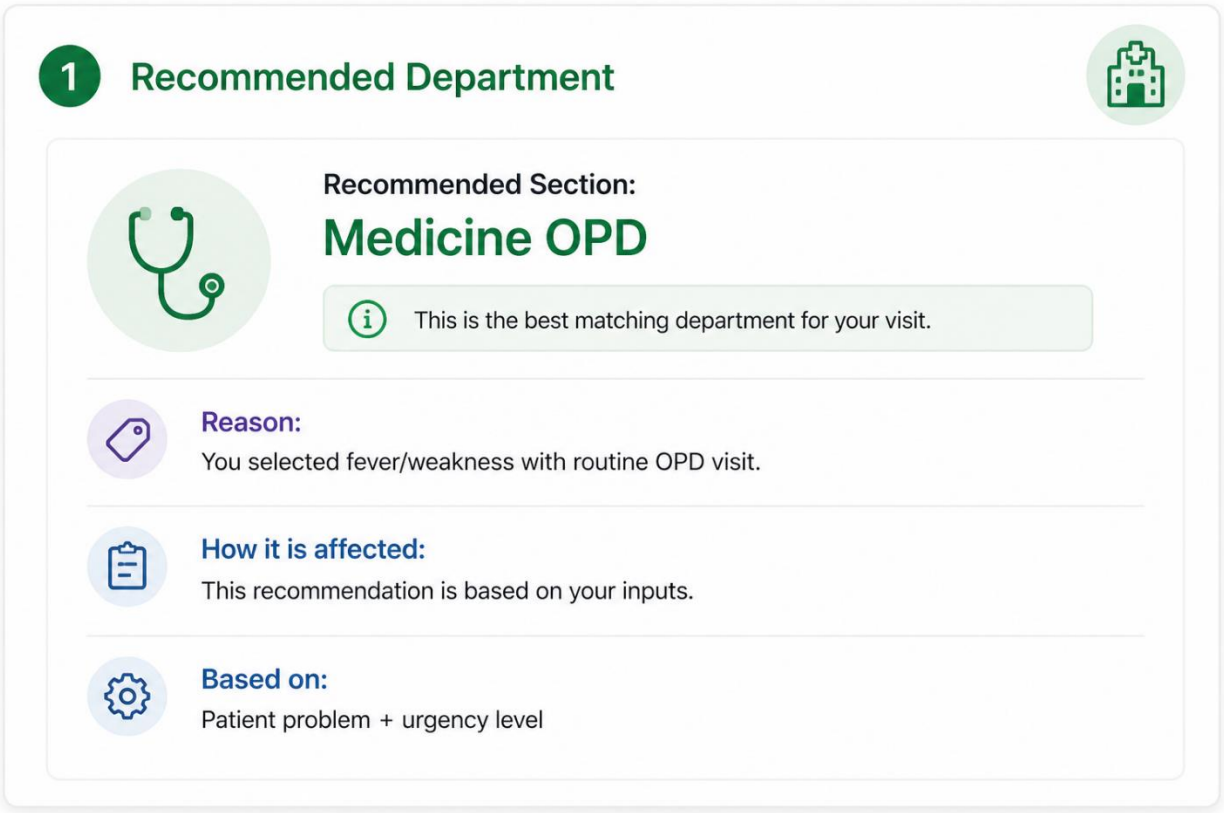
Reason

Reason:
You selected fever/weakness with routine OPD visit.

How it is affected

Based on:

Patient problem + urgency level



Result block 2: Suggested route

Example

Suggested Route:
Main Gate → Reception → Token Counter → Medicine OPD → Pharmacy → Exit

How it is affected

Based on:

Department + token status + documents + patient type

Route examples

If documents complete and token already taken

Main Gate → Reception → Medicine OPD → Pharmacy → Exit

If token needed

Main Gate → Reception → Token Counter → Medicine OPD → Pharmacy → Exit

If documents incomplete

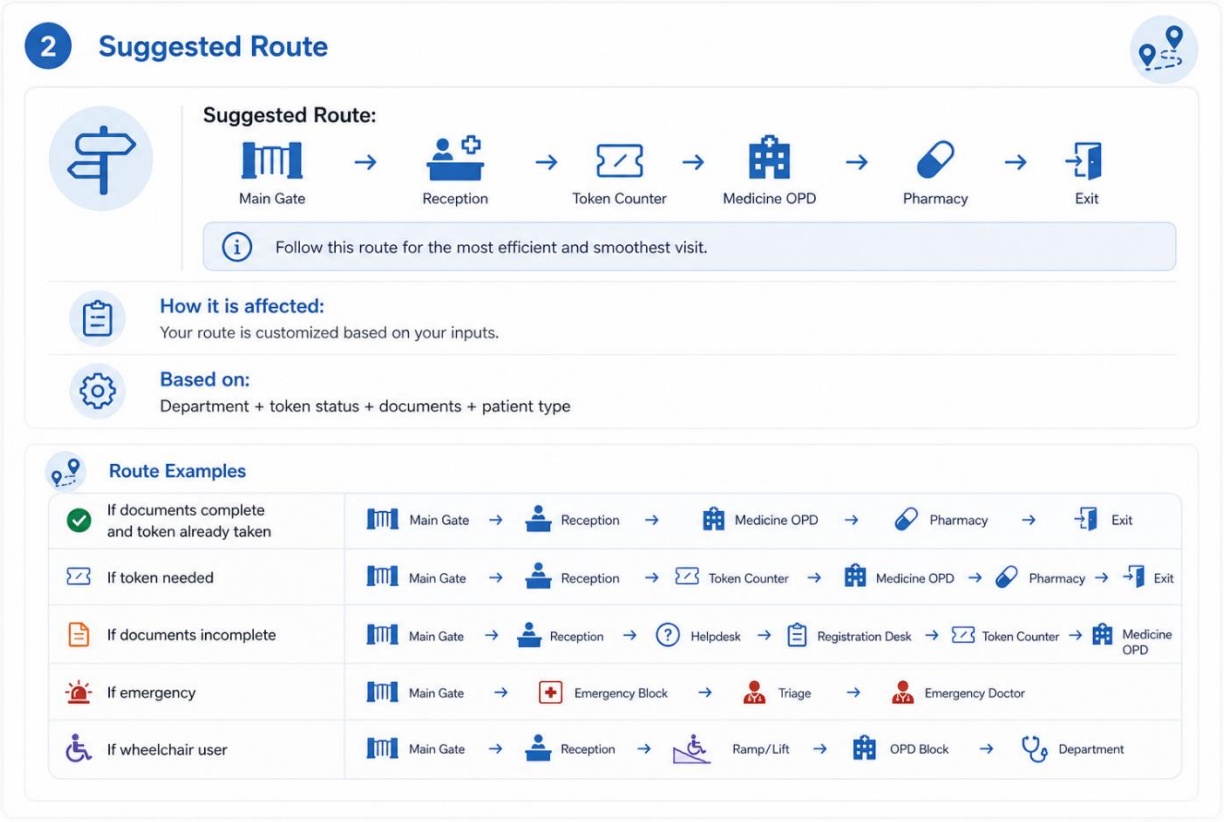
Main Gate → Reception → Helpdesk → Registration Desk → Token Counter → Medicine OPD

If emergency

Main Gate → Emergency Block → Triage → Emergency Doctor

If wheelchair user

Main Gate → Reception → Ramp/Lift → OPD Block → Department



Result block 3: Doctor availability

Example

Doctor Availability:
Medium risk

OPD doctors are usually available during morning hours.
Your selected arrival time is late, so availability risk is increased.

How it is affected

Based on:

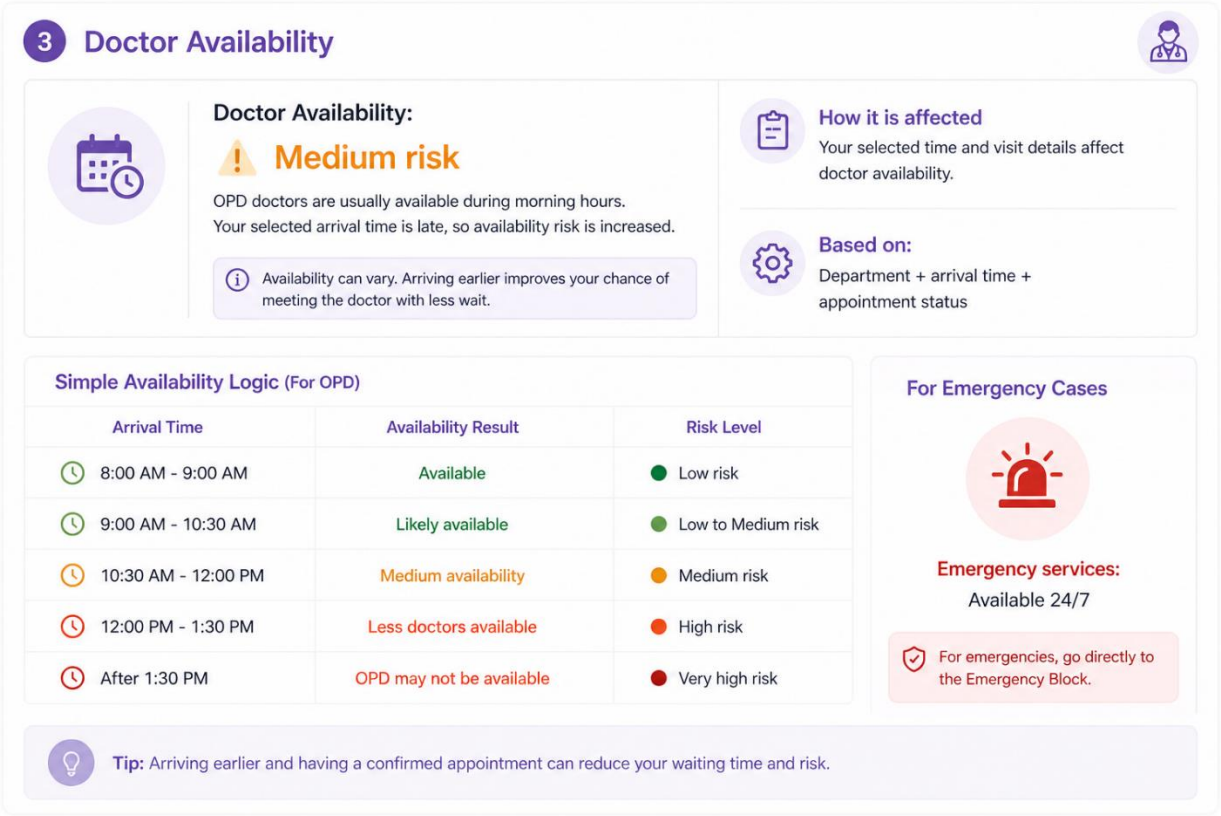
Department + arrival time + appointment status

Simple availability logic

Arrival time	Availability result
8:00 AM - 9:00 AM	Available / low risk
9:00 AM - 10:30 AM	Likely available
10:30 AM - 12:00 PM	Medium risk
12:00 PM - 1:30 PM	High risk
After 1:30 PM	OPD may not be available

For emergency:

Emergency services: Available 24/7



Result block 4: Estimated waiting time

Example

Estimated Waiting Time:
2 hours 15 minutes

How it is affected

Based on:

arrival time + appointment + token + documents + department crowd

Simple waiting-time formula

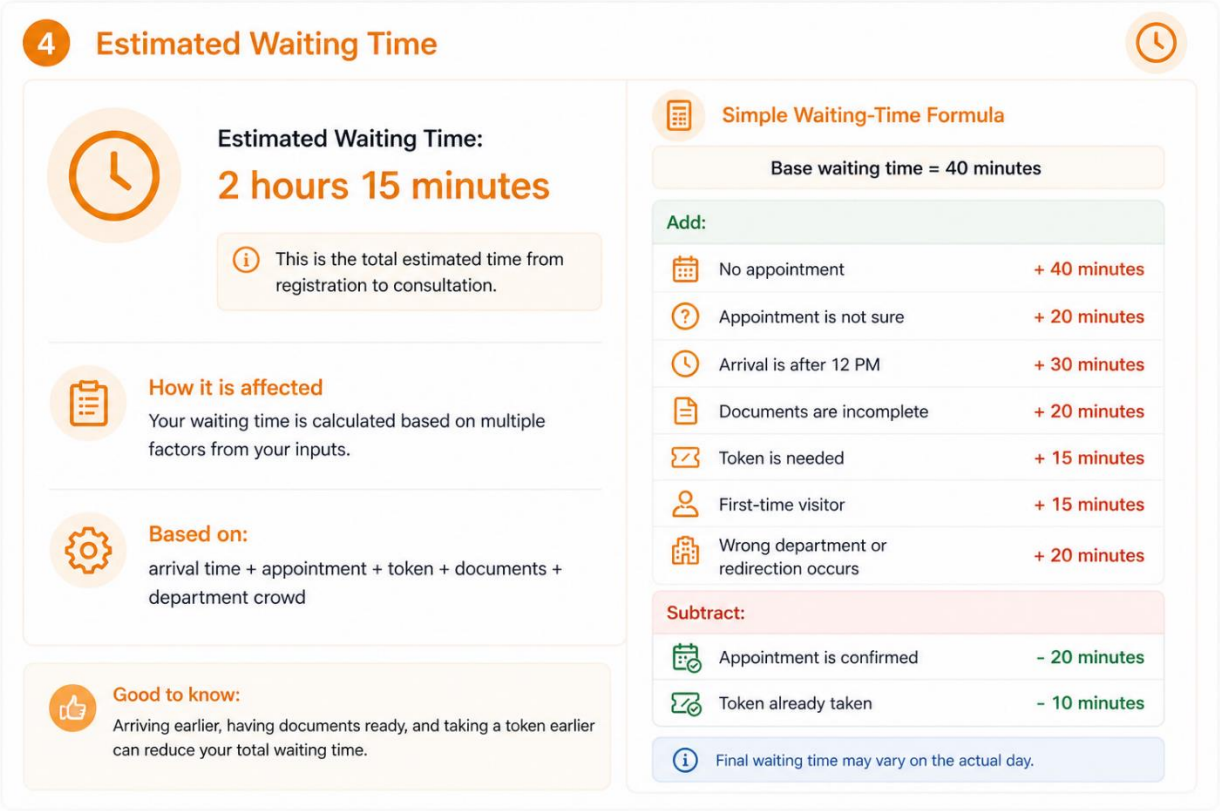
Use this in your assignment report:

Base waiting time = 40 minutes

Add:
+ 40 minutes if no appointment
+ 20 minutes if appointment is not sure

+ 30 minutes if arrival is after 12 PM
+ 20 minutes if documents are incomplete
+ 15 minutes if token is needed
+ 15 minutes if first-time visitor
+ 20 minutes if wrong department or redirection occurs

Subtract:
- 20 minutes if appointment is confirmed
- 10 minutes if token already taken



Result block 5: Risk and stress indicators

Showed as bars.

Example

Waiting Time: High
Walking Effort: High
Missed Doctor Risk: Medium
Stress Level: High
Visit Success: 60%

How each effect is calculated

Effect	**Depends on**
Waiting time	arrival, appointment, token, documents
Walking effort	route length, patient type
Missed doctor risk	arrival time, appointment
Stress level	waiting time, confusion, documents, first visit
Visit success	correct department, arrival time, documents



Result block 6: Persuasive insight

This is very important.

The simulation should tell the user **why** the result happened.

Example bad-planning result

Persuasive Insight:
Your visit became longer because you arrived late, had no appointment, and your documents were incomplete.

Planning before arrival could reduce waiting time and stress.

Example good-planning result

Persuasive Insight:
Your visit plan is efficient. Arriving early with complete documents and appointment reduces waiting time and missed doctor risk.

Example emergency result

Persuasive Insight:
For emergency symptoms, going directly to Emergency is safer than waiting in OPD.

